



Vibrio Cholerae — MCQ Practice

Microbiology GIT Block · MEDucation by Mattin Majid

HOW TO USE THIS SET

20 questions, 5 options each (A–E). Answer on paper or in your head, then check every answer against the key on the final page.

4 Easy · 10 Medium · 6 Hard · suggested time ≈ 25 minutes.

Q Questions

1. What is the characteristic morphology of *Vibrio cholerae* on Gram stain?

EASY

- A. Gram-positive coccus in chains
- B. Gram-negative diplococcus
- C. Gram-negative, curved (comma-shaped) rod
- D. Acid-fast branching rod
- E. Gram-positive spore-forming rod

2. Which enzyme test is positive for *Vibrio cholerae* and helps distinguish it from *Enterobacteriaceae*?

EASY

- A. Oxidase
- B. Catalase
- C. Coagulase
- D. Urease
- E. Indole

3. What is the term used for the classic non-bloody, watery stool of cholera?

EASY

- A. Currant-jelly stool
- B. Rice-water stool
- C. Pea-soup stool
- D. Bloody mucoid stool
- E. Steatorrheic stool

4. What is the untreated mortality rate of cholera?

EASY

- A. About 1%
- B. About 5%
- C. About 90%
- D. About 40%
- E. About 15%

5. Which biotype of *V. cholerae* O1 is responsible for the current global pandemic?

MEDIUM

- A. Classic
- B. Ogawa
- C. El Tor
- D. Inaba
- E. Hikojima

6. Which two features distinguish a biotype from a serotype of *V. cholerae*?

MEDIUM

- A. Biotype = antigenic differences; serotype = biochemical reactions
- B. Biotype = biochemical reactions; serotype = antigenic differences
- C. Both refer to antigenic differences only
- D. Both refer to biochemical reactions only
- E. Biotype = capsule type; serotype = flagellar type

7. Approximately how many *V. cholerae* organisms must be ingested via water to cause infection in a person with normal gastric acidity?

MEDIUM

- A. About 10^2
- B. About 10^4
- C. About 10^6
- D. About 10^1
- E. About 10^{10}

8. Why is a much smaller inoculum of *V. cholerae* sufficient to cause disease when transmitted via food rather than water?

MEDIUM

- A. Food organisms are a more virulent strain
- B. Food increases intestinal motility
- C. Food organisms are non-toxigenic and act by different mechanism
- D. Food buffers gastric acid, protecting the organisms
- E. Food raises gastric pH permanently

9. Which climatic conditions favor *Vibrio cholerae* survival and transmission?

MEDIUM

- A. Cold, acidic, dry conditions
- B. High altitude, low humidity
- C. Freezing temperatures
- D. Strongly acidic soil
- E. Warm, monsoon, alkaline, saline conditions

10. The 2010 cholera outbreak in Haiti was linked to which event?

MEDIUM

- A. Introduction by relief workers after a major earthquake
- B. A contaminated municipal water treatment plant built decades earlier
- C. A local seafood market outbreak
- D. Importation via untreated sewage from a cruise ship
- E. Zoonotic transmission from livestock

11. Rita Colwell's research found that *V. cholerae* bacteria commonly attach to which organism in water?

MEDIUM

- A. Freshwater snails
- B. Aquatic insect larvae
- C. Algae blooms only
- D. Fish gills
- E. Copepods (plankton)

12. Filtering water through folded sari cloth, as studied by Rita Colwell, achieved what approximate reduction in bacteria and cholera cases?

MEDIUM

- A. 99% reduction in bacteria, 50% reduction in cholera cases
- B. 50% reduction in bacteria, 99% reduction in cases
- C. 10% reduction in bacteria, 10% reduction in cases
- D. No measurable effect
- E. 100% elimination of both

13. Which receptor does the B subunit of cholera toxin bind on the intestinal epithelial cell?

MEDIUM

- A. CFTR chloride channel directly
- B. Ganglioside GM1
- C. Toll-like receptor 4
- D. ACE2 receptor
- E. Sodium-glucose cotransporter

14. What is the direct biochemical consequence of the cholera toxin A subunit's action on Gsα?

HARD

- A. Inhibition of adenylate cyclase, lowering cAMP
- B. Direct opening of calcium channels
- C. Inhibition of the sodium-potassium ATPase
- D. Constant activation of adenylate cyclase, raising intracellular cAMP
- E. Activation of guanylate cyclase and cGMP

15. On MacConkey agar, what colony appearance is typical of *Vibrio cholerae*?

MEDIUM

- A. Pink lactose-fermenting colonies
- B. Black colonies with metallic sheen
- C. Green colonies with dark centers
- D. Colorless colonies (lactose fermented slowly)
- E. Mucoid pink colonies

16. On TSI agar, what reaction pattern is typical for *Vibrio cholerae*?

HARD

- A. Alkaline slant and acid butt with gas
- B. Acid slant and alkaline butt with H₂S
- C. Acid slant and acid butt, without gas or H₂S
- D. Alkaline slant and alkaline butt
- E. Acid slant and acid butt with abundant gas

17. Which patient population should receive zinc supplementation for suspected cholera, per the guidance given?

HARD

- A. All adult patients regardless of age
- B. Children 6 months to 5 years
- C. Only pregnant women
- D. Only immunocompromised adults
- E. Neonates under 1 month

18. Which oral cholera vaccine is available in the United States for travelers to O1-endemic areas?

HARD

- A. A killed whole-cell injectable vaccine only
- B. An mRNA-based vaccine
- C. Vaxchora (live attenuated)
- D. A conjugate polysaccharide vaccine
- E. No oral vaccine is available in the US

19. An 18-year-old woman returning from Ecuador develops profuse rice-water diarrhea then vomiting, no fever, BP 88/50, dry mucous membranes; Wright stain of stool shows no leukocytes and stool is occult-blood negative. Which organism is most likely?

HARD

- A. *Escherichia coli* O157:H7
- B. *Salmonella typhi*
- C. *Clostridium difficile*
- D. *Shigella sonnei*
- E. *Vibrio cholerae*

20. Why do non-cholera *Vibrio* infections (vibrioses) such as *V. vulnificus* and *V. parahaemolyticus* appear to be increasing?

HARD

- A. Increased raw oyster consumption, increased diagnostic awareness, and a warming climate expanding *Vibrio* habitat
- B. A new highly virulent global clone has emerged
- C. Universal loss of gastric acid suppression therapy
- D. Mandatory vaccination has paradoxically increased susceptibility
- E. Decreased seafood consumption has selected for resistant strains

✓ Answer key

Q	Answer	Difficulty	Why
1	C	EASY	<i>V. cholerae</i> is a Gram-negative, curved/comma-shaped bacillus with a single polar flagellum.
2	A	EASY	<i>V. cholerae</i> is oxidase-positive, unlike members of the Enterobacteriaceae.
3	B	EASY	Rice-water stool describes the pale, watery, mucus-flecked stool typical of cholera.
4	D	EASY	The mortality rate without treatment is stated as 40%, reflecting rapid, severe fluid loss.
5	C	MEDIUM	O1 has two biotypes, Classic and El Tor; El Tor is responsible for the current global pandemic.
6	B	MEDIUM	Biotypes are distinguished by biochemical reactions, serotypes by antigenic (O-antigen) differences.
7	E	MEDIUM	Because organisms are acid-susceptible, a large inoculum ($\sim 10^{10}$) is needed via water, versus 10^2 – 10^4 via food.
8	D	MEDIUM	The buffering capacity of food protects vibrios from gastric acid, lowering the infective dose to 10^2 – 10^4 .
9	E	MEDIUM	Warm, monsoon, alkaline and saline conditions favor <i>Vibrio</i> migration and survival; cold/acidic/dry inhibits it.
10	A	MEDIUM	Cholera was unintentionally introduced to Haiti by relief workers after the catastrophic earthquake, causing >770,000 infections.
11	E	MEDIUM	She discovered that most cholera bacteria attach to plankton called copepods, informing the sari-cloth filtration strategy.
12	A	MEDIUM	Sari-cloth filtration removed about 99% of bacteria and cut cholera cases by roughly 50%.
13	B	MEDIUM	GM1 ganglioside is the mucosal receptor for the B subunit, which promotes entry of the A subunit.
14	D	HARD	ADP-ribosylation locks G_{α} in the active state, causing constant adenylate cyclase stimulation and cAMP overproduction.
15	D	MEDIUM	<i>V. cholerae</i> ferments lactose slowly, producing colorless colonies on MacConkey agar.
16	C	HARD	<i>V. cholerae</i> ferments sucrose, giving an acid slant/acid butt without gas or H_2S production.
17	B	HARD	Children 6 months to 5 years with suspected cholera should receive zinc when available, to reduce severity/duration.
18	C	HARD	Vaxchora is an oral, live attenuated vaccine available in the US for travelers to areas where O1 cholera is endemic.
19	E	HARD	A secretory, noninflammatory diarrhea with no fecal leukocytes or blood, causing profound dehydration, is classic for <i>V. cholerae</i> ; the other organisms cause inflammatory or febrile illness with fecal leukocytes.
20	A	HARD	The lecture attributes the apparent rise to more raw oyster consumption, more awareness/diagnosis, and climate warming expanding the bacteria's aquatic habitat.